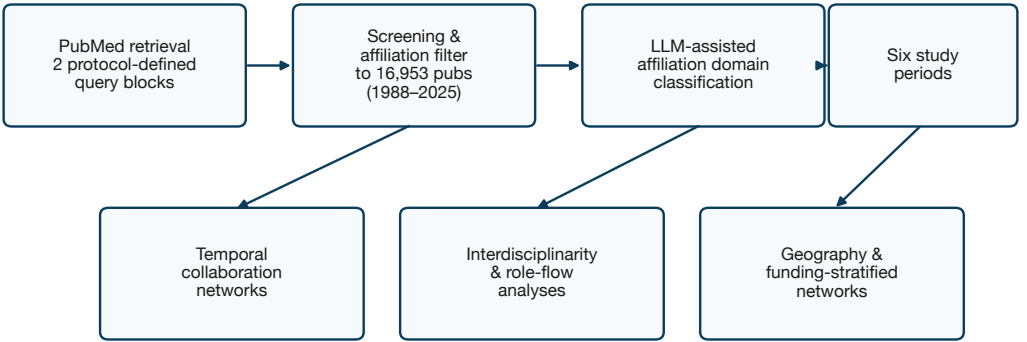
PubMed retrieval
2 protocol-defined
query blocks
Screening &
affiliation filter
to 16,953 pubs
(1988–2025)
LLM-assisted
affiliation domain
classification
Six study
periods
Temporal
collaboration
networks
Interdisciplinarity
& role-flow
analyses
Geography &
funding-stratified
networks